

Lecture 4: Proof Notes for Board Work

ISE 5406 – Optimization Theory

These notes contain the proofs to write on the board during lecture.

Green = Steps **Orange** = Key terms **Purple** = Results/Conclusions

1. Convexity of $f(x) = x^2$ from Definition

Statement

Prove that $f(x) = x^2$ is **convex** on $\mathbb{R}$ using the definition.

Step 1: State what we need to show

For all $x_1, x_2 \in \mathbb{R}$ and $\lambda \in [0, 1]$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

Step 2: Compute the LHS

$$\text{LHS} = (\lambda x_1 + (1 - \lambda)x_2)^2$$

$$\text{Expand: } = \lambda^2 x_1^2 + 2\lambda(1 - \lambda)x_1 x_2 + (1 - \lambda)^2 x_2^2$$

Step 3: Compute the RHS

$$\text{RHS} = \lambda x_1^2 + (1 - \lambda)x_2^2$$

Step 4: Compute RHS – LHS

$$\text{RHS} - \text{LHS} =$$

$$\lambda x_1^2 + (1 - \lambda)x_2^2 - \lambda^2 x_1^2 - 2\lambda(1 - \lambda)x_1 x_2 - (1 - \lambda)^2 x_2^2$$

[Factor out $\lambda(1 - \lambda)$ from each term]

$$= \lambda(1 - \lambda)x_1^2 - 2\lambda(1 - \lambda)x_1 x_2 + \lambda(1 - \lambda)x_2^2$$

$$= \lambda(1 - \lambda)(x_1^2 - 2x_1 x_2 + x_2^2)$$

$$= \lambda(1 - \lambda)(x_1 - x_2)^2$$

Step 5: Show this is nonnegative

Since $\lambda \in [0, 1]$: we have $\lambda \geq 0$ and $(1 - \lambda) \geq 0$

$$\Rightarrow \lambda(1 - \lambda) \geq 0$$

Also $(x_1 - x_2)^2 \geq 0$ always

$$\Rightarrow \text{RHS} - \text{LHS} = \lambda(1 - \lambda)(x_1 - x_2)^2 \geq 0$$

Therefore $f(x) = x^2$ is convex $\square$

2. Running Example: Convexity via Composition Rules

Statement

Prove that $f(x, y) = x^2 + 2y^2$ is **convex** on $\mathbb{R}^2$ using composition rules.

Step 1: Recall what we know

We just proved: $g(t) = t^2$ is convex on $\mathbb{R}$

Step 2: Identify the building blocks

Define $g_1(x, y) = x^2$ and $g_2(x, y) = y^2$

Each is a composition: t^2 composed with a linear projection

[Linear functions are convex, t^2 is convex, composition with affine preserves convexity]

$\Rightarrow$ g_1 and g_2 are both **convex**

Step 3: Apply preservation rule

$f(x, y) = 1 \cdot g_1(x, y) + 2 \cdot g_2(x, y)$

This is a **nonnegative weighted sum** (weights $1 \geq 0$ and $2 \geq 0$)

By preservation rule: nonnegative sums of convex functions are convex

Therefore $f(x, y) = x^2 + 2y^2$ is convex $\square$

*[Note: Since x^2 and y^2 are strictly convex with positive weights, the sum is also **strictly convex**]*

3. First-Order Condition for Convexity

Statement

Let $S \subseteq \mathbb{R}^n$ be open and convex, $f : S \rightarrow \mathbb{R}$ differentiable. Then f is **convex** on S if and only if:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in S$$

Step 1: $(\Rightarrow)$ Assume f is convex. Show first-order condition holds.

Let $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in (0, 1]$

By **convexity**:

$$f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) \leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$$

$$[\text{Note: } \mathbf{x} + \lambda(\mathbf{y} - \mathbf{x}) = (1 - \lambda)\mathbf{x} + \lambda\mathbf{y}]$$

Step 2: Rearrange the inequality

$$f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x}) \leq \lambda(f(\mathbf{y}) - f(\mathbf{x}))$$

Divide by $\lambda > 0$:

$$\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x})$$

Step 3: Take limit as $\lambda \rightarrow 0^+$

$$\text{LHS} \rightarrow \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \quad [\text{by definition of directional derivative}]$$

$$\Rightarrow \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x})$$

$$\boxed{\text{Rearranging: } f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \quad \checkmark}$$

Step 4: $(\Leftarrow)$ Assume first-order condition. Show f is convex.

Let $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$

Define $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$

$$[\text{We want to show } f(\mathbf{z}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})]$$

Step 5: Apply first-order condition twice

Apply at $\mathbf{z}$ with point $\mathbf{x}$:

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^T(\mathbf{x} - \mathbf{z})$$

Apply at $\mathbf{z}$ with point $\mathbf{y}$:

$$f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^T(\mathbf{y} - \mathbf{z})$$

Step 6: Combine with weights

Multiply first by λ , second by $(1 - \lambda)$:

$$\lambda f(\mathbf{x}) \geq \lambda f(\mathbf{z}) + \lambda \nabla f(\mathbf{z})^T (\mathbf{x} - \mathbf{z})$$

$$(1 - \lambda)f(\mathbf{y}) \geq (1 - \lambda)f(\mathbf{z}) + (1 - \lambda)\nabla f(\mathbf{z})^T (\mathbf{y} - \mathbf{z})$$

Add:

$$\lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^T [\lambda(\mathbf{x} - \mathbf{z}) + (1 - \lambda)(\mathbf{y} - \mathbf{z})]$$

Step 7: Simplify the bracket

$$\lambda(\mathbf{x} - \mathbf{z}) + (1 - \lambda)(\mathbf{y} - \mathbf{z}) = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y} - \mathbf{z} = \mathbf{z} - \mathbf{z} = \mathbf{0}$$

$$\Rightarrow \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \geq f(\mathbf{z})$$

This is exactly convexity of f $\square$

4. Running Example: First-Order Proof

Statement

Prove that $f(x, y) = x^2 + 2y^2$ is **convex** on $\mathbb{R}^2$ using the first-order condition.

Step 1: Compute the gradient

$$f(x, y) = x^2 + 2y^2$$

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: State the first-order condition

Need to verify: for all $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$:

$$f(x_2, y_2) \geq f(x_1, y_1) + \nabla f(x_1, y_1)^T \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Step 3: Expand the LHS

$$\text{LHS} = f(x_2, y_2) = x_2^2 + 2y_2^2$$

Step 4: Expand the RHS

$$\text{RHS} = x_1^2 + 2y_1^2 + \begin{pmatrix} 2x_1 \\ 4y_1 \end{pmatrix}^T \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

$$\begin{aligned} &= x_1^2 + 2y_1^2 + 2x_1(x_2 - x_1) + 4y_1(y_2 - y_1) \\ &= x_1^2 + 2y_1^2 + 2x_1x_2 - 2x_1^2 + 4y_1y_2 - 4y_1^2 \\ &= -x_1^2 + 2x_1x_2 - 2y_1^2 + 4y_1y_2 \end{aligned}$$

Step 5: Compute LHS – RHS

$$\text{LHS} - \text{RHS} = x_2^2 + 2y_2^2 - (-x_1^2 + 2x_1x_2 - 2y_1^2 + 4y_1y_2)$$

$$\begin{aligned} &= x_2^2 + x_1^2 - 2x_1x_2 + 2y_2^2 + 2y_1^2 - 4y_1y_2 \\ &= (x_2 - x_1)^2 + 2(y_2 - y_1)^2 \end{aligned}$$

Step 6: Verify nonnegativity

$$(x_2 - x_1)^2 \geq 0 \text{ always}$$

$$(y_2 - y_1)^2 \geq 0 \text{ always}$$

$$\Rightarrow (x_2 - x_1)^2 + 2(y_2 - y_1)^2 \geq 0$$

First-order condition holds, so $f(x, y) = x^2 + 2y^2$ is convex $\square$

5. Second-Order Condition for Convexity

Statement

Let $S \subseteq \mathbb{R}^n$ be open and convex, $f : S \rightarrow \mathbb{R}$ twice continuously differentiable. Then f is **convex** on S if and only if:

$$\nabla^2 f(\mathbf{x}) \succeq 0 \quad \text{for all } \mathbf{x} \in S$$

Step 1: $(\Rightarrow)$ Assume f is convex. Show Hessian is PSD.

[We'll use the first-order condition which we just proved]

By first-order condition:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in S$$

Step 2: Use Taylor expansion

Let $\mathbf{y} = \mathbf{x} + t\mathbf{d}$ for small $t > 0$ and any direction $\mathbf{d}$

Taylor expansion of f around $\mathbf{x}$:

$$f(\mathbf{x} + t\mathbf{d}) = f(\mathbf{x}) + t\nabla f(\mathbf{x})^T \mathbf{d} + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(t^2)$$

Step 3: Apply first-order condition

$$\text{From first-order: } f(\mathbf{x} + t\mathbf{d}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (t\mathbf{d})$$

Substituting Taylor:

$$f(\mathbf{x}) + t\nabla f(\mathbf{x})^T \mathbf{d} + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(t^2) \geq f(\mathbf{x}) + t\nabla f(\mathbf{x})^T \mathbf{d}$$

Step 4: Simplify

Cancel common terms:

$$\frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(t^2) \geq 0$$

Divide by $\frac{t^2}{2} > 0$ and take $t \rightarrow 0$:

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} \geq 0 \quad \text{for all } \mathbf{d} \in \mathbb{R}^n$$

This means $\nabla^2 f(\mathbf{x}) \succeq 0$ (positive semidefinite) ✓

Step 5: $(\Leftarrow)$ Assume $\nabla^2 f(\mathbf{x}) \succeq 0$. Show f is convex.

[This direction uses Taylor's theorem with integral remainder – see Bazaraa Theorem 3.3.8]

The key idea: For $\mathbf{x}, \mathbf{y} \in S$, Taylor with remainder gives:

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{1}{2} (\mathbf{y} - \mathbf{x})^T \nabla^2 f(\mathbf{z}) (\mathbf{y} - \mathbf{x})$$

for some $\mathbf{z}$ on line segment $[\mathbf{x}, \mathbf{y}]$.

Since $\nabla^2 f(\mathbf{z}) \succeq 0$: last term ≥ 0

So $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$, which is first-order condition $\square$

6. Running Example: Second-Order Proof

Statement

Prove that $f(x, y) = x^2 + 2y^2$ is **strictly convex** on $\mathbb{R}^2$ using the second-order condition.

Step 1: Compute the gradient

$$f(x, y) = x^2 + 2y^2$$

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Step 2: Compute the Hessian

Second partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = 4, \quad \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$\nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

[Note: This is constant – does not depend on (x, y)]

Step 3: Check positive definiteness

(Method 1) Eigenvalues

$$\text{Eigenvalues of } \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \text{ are } \lambda_1 = 2 \text{ and } \lambda_2 = 4$$

Both eigenvalues > 0

$\Rightarrow \nabla^2 f \succ 0$ (positive definite)

(Method 2) Direct verification

For any $\mathbf{d} = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \neq \mathbf{0}$:

$$\mathbf{d}^T \nabla^2 f \mathbf{d} = \begin{pmatrix} d_1 & d_2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = 2d_1^2 + 4d_2^2 > 0$$

[Strict inequality since $\mathbf{d} \neq \mathbf{0}$]

Since $\nabla^2 f \succ 0$ everywhere, $f(x, y) = x^2 + 2y^2$ is strictly convex $\square$

7. Epigraph Characterization of Convexity

Statement

Let $S \subseteq \mathbb{R}^n$ be convex and $f : S \rightarrow \mathbb{R}$. Then f is **convex** if and only if $\text{epi}(f)$ is a **convex set**.

Step 1: $(\Rightarrow)$ Assume f is convex. Show $\text{epi}(f)$ is convex.

Let $(\mathbf{x}_1, z_1), (\mathbf{x}_2, z_2) \in \text{epi}(f)$

By definition of epigraph: $z_1 \geq f(\mathbf{x}_1)$ and $z_2 \geq f(\mathbf{x}_2)$

Let $\lambda \in [0, 1]$. Consider $\lambda(\mathbf{x}_1, z_1) + (1 - \lambda)(\mathbf{x}_2, z_2)$

This equals $(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2, \lambda z_1 + (1 - \lambda)z_2)$

Step 2: Show this point is in $\text{epi}(f)$

Need: $\lambda z_1 + (1 - \lambda)z_2 \geq f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2)$

Since $z_1 \geq f(\mathbf{x}_1)$ and $z_2 \geq f(\mathbf{x}_2)$:

$$\lambda z_1 + (1 - \lambda)z_2 \geq \lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2)$$

Since f is **convex**:

$$\lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) \geq f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2)$$

$$\Rightarrow \lambda z_1 + (1 - \lambda)z_2 \geq f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2)$$

So $(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2, \lambda z_1 + (1 - \lambda)z_2) \in \text{epi}(f)$ ✓

Step 3: $(\Leftarrow)$ Assume $\text{epi}(f)$ is convex. Show f is convex.

Let $\mathbf{x}_1, \mathbf{x}_2 \in S$ and $\lambda \in [0, 1]$

Note: $(\mathbf{x}_1, f(\mathbf{x}_1)) \in \text{epi}(f)$ and $(\mathbf{x}_2, f(\mathbf{x}_2)) \in \text{epi}(f)$

Since $\text{epi}(f)$ is **convex**:

$$\lambda(\mathbf{x}_1, f(\mathbf{x}_1)) + (1 - \lambda)(\mathbf{x}_2, f(\mathbf{x}_2)) \in \text{epi}(f)$$

This means: $(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2, \lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2)) \in \text{epi}(f)$

Step 4: Apply definition of epigraph

By definition of $\text{epi}(f)$:

$$\lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) \geq f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2)$$

This is exactly the definition of convexity of f □

Summary: Three Ways to Prove Convexity

Comparison for $f(x, y) = x^2 + 2y^2$

Method	Key Calculation	Result
Composition	Sum of convex: $x^2 + 2 \cdot y^2$	Strictly convex
First-Order	Show $(x_2 - x_1)^2 + 2(y_2 - y_1)^2 \geq 0$	Convex
Second-Order	$\nabla^2 f = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, eigenvalues 2, 4 > 0	Strictly convex

When to use each method:

- **Composition rules:** When function is built from known convex functions using sums, max, affine composition, etc.
- **First-order condition:** When you need to prove convexity and connect to optimality (gradient = 0 at minimum). Useful for proving theorems.
- **Second-order condition:** Often the quickest for smooth functions. Just compute the Hessian and check eigenvalues or use other PSD criteria.